

Generation Interconnection System Impact Study Report

For Queue Project Q22-2176

Hood 138 kV
October 15, 2024

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q22-2176. The Interconnection Customer proposes a 291.9 MW battery energy storage facility interconnecting at the Hood 138 kV substation in the Maryland area. The request was received on September 07, 2022. The study evaluated system intact and single contingency conditions using the regional planning case, including 430 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 26 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$53,869,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Oliver St to Bethesda 138 KV Line	Grant Avenue to Cahokia 138 KV	223.7	200	100.0
Doucette Substation to Bethesda 138 KV Line	Cahokia to Grant Avenue 138 KV	216.0	248	100.0
Grant Avenue to Cahokia 138 KV line	Doucette to Bethesda 138 KV	214.1	250	100.0
Hood to Cahokia 138 KV Line	Doucette Substation to Bethesda 138 KV line	208.5	241	100.0
Oliver St to Hood 138 KV	Grant Avenue to Cahokia 138 KV Line	191.3	216	100.0
Doucette to New Sterling 138 KV	Cahokia to Grant Avenue 138 KV Line	174.6	225	73.6
Twinbrook to Kensington 138 KV Cir. 1	Twinbrook to Kensington Substation 138 KV	147.0	287	6.5
Twinbrook to Kensngton 138 KV	Twinbrook to Kensington Substation 138 KV Line 3	147.0	287	6.5
Twinbrook to Kensington 138 KV Ckt 3	Twinbrook to Kensington 138 KV	147.0	287	6.5
Flanagan to Linden 138 KV	Linden to Flanagan 138 KV Circuit 2	142.1	256	18.8
Flanagan to Linden 138 KV Circuit 2	Flanagan Substation to Linden 138 KV	142.1	256	18.8
Swinks Mill to Shepard 138 KV Line	Swinks Mill to Shepard Substation 138 KV Circuit 2	134.8	392	7.9
Bloomington to Shepard 138 KV Cir. 1	Bloomington to Shepard 138 KV L	128.2	387	7.9
Shepard to Swinks Mill 138 KV	Swinks Mill to Shepard 138 KV Circuit 1	126.9	416	7.9
Tokoma to Grant Avenue 138 KV L	Grant Avenue to Seek Lane 138 KV	125.5	242	55.2
Flanagan to Seek Lane 138 KV	Colesville to Darien 138 KV	124.9	422	20.3

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Seek Lane to Grant Avenue 138 KV	Bethesda to Doucette 138 KV	122.9	305	45.5
Swinks Mill to Carderock 138 KV #2	Swinks Mill to Carderock 138 KV Cir. 1	122.5	441	7.7
Tillatoba Substation to Twinbrook 138 KV Line	Flanagan to Seek Lane 138 KV	121.4	252	5.7
Bloomington to Shepard 138 KV #2	Shepard to Bloomington 138 KV	120.0	414	7.9
Swinks Mill to Carderock 138 KV Line 1	Swinks Mill to Carderock 138 KV #2	119.6	451	7.7
Kensington to Nih 138 KV	Tinbrook to Kensington 138 KV Line 2	118.9	220	5.8
Sobieski to Carrollton-Josey Lane 138 KV	Flanagan to Seek Lane 138 KV	116.6	234	9.9
Linden to Kensington 138 KV Cir. 1	Linden to Kensington 138 KV Line 2	115.0	534	11.0
Kensington to Linden 138 KV	Linden to Kensington 138 KV Ckt 1	115.0	534	11.0
Bald Mtn to Summit Lake 138 KV Line	Darien to Colesville 138 KV	104.1	194	6.3

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Oliver St to Bethesda 138 KV Line	Rebuild line with higher-capacity conductor	\$1,329,000	100% (\$1,329,000)
Bethesda to Doucette 138 KV line	Rebuild line with higher-capacity conductor	\$1,328,000	100% (\$1,328,000)
Grant Avenue Substation to Cahokia 138 KV	Rebuild line with higher-capacity conductor	\$927,000	100% (\$927,000)
Hood Substation to Caokia 138 KV	Rebuild line with higher-capacity conductor	\$2,275,000	100% (\$2,275,000)
Oliver St to Hood 138 KV	Rebuild line with higher-capacity conductor	\$1,989,000	100% (\$1,989,000)
Doucette to New Sterling 138 KV	Rebuild line with higher-capacity conductor	\$2,888,000	100% (\$2,888,000)
Twinbrook to Kensington 138 KV Cir. 1	Rebuild line with higher-capacity conductor	\$6,017,000	100% (\$6,017,000)
Kensington to Twinbrook 138 KV #2	Rebuild line with higher-capacity conductor	\$5,850,000	100% (\$5,850,000)
Twinbrook to Kensington 138 KV Ckt 3	Rebuild line with higher-capacity conductor	\$5,693,000	100% (\$5,693,000)
Flanagan to Linden 138 KV line	Rebuild line with higher-capacity conductor	\$2,909,000	100% (\$2,909,000)
Flanagan to Linden 138 KV Circuit 2	Rebuild line with higher-capacity conductor	\$4,054,000	100% (\$4,054,000)
Swinks Mill Substation to Shepard 138 KV	Rebuild line with higher-capacity conductor	\$716,000	63% (\$453,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Bloomington to Shepard 138 KV Cir. 1	Rebuild line with higher-capacity conductor	\$871,000	63% (\$551,000)
Swinks Mill to Shepard 138 KV #2	Rebuild line with higher-capacity conductor	\$814,000	63% (\$514,000)
Tokoma to Grant Avenue 138 KV	Rebuild line with higher-capacity conductor	\$1,673,000	100% (\$1,673,000)
Flanagan to Seek Lane 138 KV line	Rebuild line with higher-capacity conductor	\$804,000	79% (\$636,000)
Grant Avenue to Seek Lane 138 KV L	Rebuild line with higher-capacity conductor	\$1,201,000	100% (\$1,201,000)
Swinks Mill to Carderock 138 KV #2	Rebuild line with higher-capacity conductor	\$4,869,000	63% (\$3,078,000)
Tillatoba to Twinbrook 138 KV	Rebuild line with higher-capacity conductor	\$1,577,000	100% (\$1,577,000)
Shepard to Bloomington 138 KV Line 2	Rebuild line with higher-capacity conductor	\$1,843,000	63% (\$1,165,000)
Swinks Mill to Carderock 138 KV Line	Rebuild line with higher-capacity conductor	\$1,856,000	63% (\$1,173,000)
Kensington to Nih 138 KV	Rebuild line with higher-capacity conductor	\$2,267,000	100% (\$2,267,000)
Carrollton-Josey Lane to Sobieski 138 KV Line	Rebuild line with higher-capacity conductor	\$1,114,000	69% (\$767,000)
Kensington to Linden 138 KV Line 1	Rebuild line with higher-capacity conductor	\$1,393,000	100% (\$1,393,000)
Kensington to Linden 138 KV	Rebuild line with higher-capacity conductor	\$1,659,000	100% (\$1,659,000)
Bald Mtn to Summit Lake 138 KV Line	Reconductor line	\$503,000	100% (\$503,000)
	Total allocated to project		\$53,869,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.